

Leonard Ndwiga

Indwiga | [Linkedin](#)

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Location: Glasgow, UK

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Mobile: +447425237540

EDUCATION

University of Nairobi

Master of Science in Bioinformatics

Nairobi, Kenya

December 2018

University of Nairobi

Bachelor of Science in Biochemistry

Nairobi, Kenya

December 2014

University of Washington

Certificate: Project Management in Global Health

United States of America

September 2023

SKILLS SUMMARY

Programming Languages: R, Python, Linux, Bash, Snakemake

Tools:

Excel, PowerPoint, R Markdown, CLC Genomics, GitHub, CZID, Singularity/Apptainer

Bioinformatics Projects:

SARS-CoV-2 Genomic analysis, Metagenomics (both DNA & RNA), Bulk Transcriptomics, *Plasmodium falciparum* Amplicon deep sequencing (both 16S rRNA and other targeted regions), Whole genome assembly: *P. falciparum* & Bacteria (reference & de-novo), ML for Laboratory Data Prediction, Sanger Sequence assembly, Iterative pipeline development.

Soft skills:

Training and Mentorship, Problem solving, Attention to Detail, Prioritization, Continuous Learning, Excellent Communication

WORK EXPERIENCE

Research Assistant | University of Glasgow – Centre for Virus Research – 27th

February 2024 – Present

- Develop and maintain bioinformatics pipelines for analysis of whole genome sequencing, metagenomics, and transcriptomics data using Bash, R, Python, and Snakemake for routine workflow automation.
- Planning, preparing, and quality controlling HTS libraries, including sequencing and result interpretation.
- Designing and performing experiments to evaluate new NGS protocols.
- Contribute to troubleshooting HTS instruments and workflows and drive the development of advanced HTS technologies.
- Document research data, maintain records and databases, draft reports and papers, and enhance research profile and reputation.
- Present work at conferences, seminars, and workshops to promote research.

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- Identify potential funding sources and help develop proposals to secure research funding.

Head of Sequencing Facility | KEMRI-Wellcome Trust Research Programme – 1st

October 2021 – 16th February 2024

- Stakeholder Communication and Leadership: Led collaboration with county health officials, external partners, and media teams to ensure the successful execution of research projects, including public awareness initiatives in genomic sequencing.
- Project Coordination and Management: Oversaw Sanger and next generation sequencing services, maintaining project timelines and quality using project management tools like Gantt charts.
- Led the SARS-CoV-2 sequencing team where I oversaw: RNA extraction, library preparation, and downstream analysis. Developed and implemented custom bioinformatics pipelines for genome assembly, variant calling, and lineage classification. Actively contributed to drafting policy briefs that guided real-time public health decision-making.
- Led multiple data analysis projects involving bash scripting, R programming, and high - throughput sequencing data, including variant analysis and phylogenetics. These contributions supported peer-reviewed publications and informed epidemiological insights.
- Training and Education: Developed and led sequencing and bioinformatics training programs for various stakeholders, including international organizations like Africa-CDC and Chan Zuckerberg Initiative.
- Documentation and Reporting: Produced comprehensive reports, policy briefs, manuscripts, and clear documentation of sequencing and research activities.
- Technical Expertise: Optimized sequencing protocols for various pathogens: parasites, bacteria and viral, developed COVID diagnostic kits, and ensured accuracy in diagnostic testing.

Research Officer | KEMRI-Wellcome Trust Research Programme - 8th October 2018 – 30th September 2021

- Performed DNA extraction and genotyping of antimalarial drug resistance and polymorphic markers using PCR and amplicon deep sequencing, with downstream variant calling and analysis to support drug efficacy trials.
- Led bioinformatics workflows for whole genome and amplicon sequencing, including QC, alignment (BWA), and variant calling (GATK/Samtools); also utilized CLC Genomics Workbench for Sanger sequence analysis to support targeted genotyping and validation.
- Co-ordinated field and lab activities in the Open Guideline Project, including qPCR screening, sample logistics, and integration of molecular and genomic data.
- Developed SOPs, managed lab procurement, and ensured uninterrupted operations through effective budgeting and forecasting.
- Contributed to COVID-19 molecular diagnostics and rapid reporting during national response efforts.

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RESEARCH GRANTS

Royal Society of Tropical Medicine and Hygiene (RSTMH) | NIHR small Grant | £5000

Project title: Genotyping Novel Plasmodium falciparum drug resistance markers in samples obtained from drug efficacy studies in Kenya

Start: 1st September 2019

End: 31st May 2021

Initiative to Develop African Research Leaders (IDeAL) | £4100

Project title: Molecular Analysis of Antimalarial Resistance Markers in Parasite Samples Obtained From Children Recruited Into a Drug Efficacy Trial in Kwale, Kenya, 2013

Start: 1st September 2016

End: 31st August 2017

LIST OF PUBLICATIONS

1. Epidemiology of COVID-19 infections on routine polymerase chain reaction (PCR) and serology testing in Coastal Kenya James Nyagwange*, **Ndwiga L.*** et. al. Feb 2022 - **equal contribution**
2. A review of the frequencies of Plasmodium falciparum Kelch 13 artemisinin resistance mutations in Africa **Ndwiga L.** et. al., 2021
3. The Plasmodium falciparum Rh5 invasion protein complex reveals an excess of rare variant mutations. **Ndwiga L.** et.al., 2021
4. Targeted Amplicon deep sequencing of ama1 and mdr1 to track within-host *P. falciparum* diversity throughout treatment in a clinical drug trial Kevin Wamae* **Leonard Ndwiga*** et. al. 2022 -**equal contribution**
5. Low frequency of Plasmodium falciparum hrp2/3 deletions from symptomatic infections at a primary healthcare facility in Kilifi, Kenya: Dorcas Okanda, **Leonard Ndwiga**

Full list of publications available at:

<https://pubmed.ncbi.nlm.nih.gov/?term=Leonard+Ndwiga>

CONFERENCES & TRAININGS

Molecular and Genomic Approaches to Clinical Microbiology in Africa (Wellcome Connecting Science) – 13th – 19th September 2025 MRC Unit, The Gambia | **Role:** Trainer

Bioinformatics with Linux and Bash Scripting: From Basics to Mastery – 30th June – 11th July 2025 | Kigali, Rwanda | **Role:** Trainer

Whole Genome Sequence Analysis Training Workshop (Virtual) – April – September 2025. Kenya Institute of Primate Research (KIPRE), Nairobi, Kenya | **Role: Trainer**
Conducting a 5-month virtual training (weekly sessions) for six research scientists on the analysis of snake whole genome sequence data using their own datasets.

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Molecular Approaches to Clinical Microbiology in Africa (Wellcome Connecting Science)

– 30th August – 6th September 2024 Stellenbosch University, South Africa | **Role:** Trainer

7th Viral Bioinformatics and Genomics Training Course, CVR, Glasgow, UK – Centre for virus Research (CVR) 24th – 28th June 2024 | **Role:** Trainee

Metagenomic and Bioinformatics Training using CZ-ID (Africa-CDC and Chan Zuckerberg Biohub) 18th – 29th September 2023 | **Role:** Trainer

Molecular Approaches to Clinical Microbiology in Africa (Wellcome Connecting Science)

- 2nd – 8th September 2023 KEMRI, Kenya | **Role:** Trainer